Susquehanna

MEDICAL IMAGE CLASSIFICATION FOR EARLY DISEASE INDICATORS

USING NIH CHESTX-RAY14

DSCI 591 Capstone 1

Winter 2026

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1. THE TEAM

- Phase I Implementation Completion

COSIMO MORELLI

Technical Lead & Data Engineer

VIRGINIA MUTHARD

Domain Expert & Analytics Lead

ADAN ORDONEZ

Project Manager & Documentation Lead

2. THE PROBLEM

INTRO

Chest radiography is an important diagnostic tool in healthcare but can be subjective and time-consuming. Advances in technology and machine learning may provide a means for more objective clinical analysis and pre-primary diagnosis.

GOALS

Determine dataset properties that most strongly shape the feasibility and validity of learning early disease indicators from chest radiographs.

USERS

Potential stakeholders include clinical radiologists, hospital administrators, ML researchers, and regulatory bodies.

3. THE DATA

SOURCE

NIH ChestX-ray14

Sourced from NIH Clinical Center

COMPOSITION

112,120 labeled images from
30,805 unique patients across
14 thoracic disease labels

METADATA LABELS

Weakly-supervised

Collected using NLP of radiological reports

4. THE LABELS

PLEURAL THICKENING

Thickening of the pleural membrane surrounding the lungs

EMPHYSEMA

Characterized by destruction of alveoli impairing oxygen exchange, most commonly caused by smoking or chronic inhalation of irritants

PNEUMOTHORAX

Accumulation of gas in the thoracic cavity, causing pain and shortness of breath

ATELECTASIS

Atelectasis is the collapse of lung tissue, typically caused by external pleural pressure or airway obstruction

MASS

Discrete tissue opacity exceeding 30 mm in diameter on chest radiograph

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NODULE

Similar to a mass but measures less than 30 mm in diameter and may present as a solitary or multiple lesions

EFFUSION

Accumulation of fluid in the extra-pleural space, creating inward pressure on the lung that can ultimately lead to collapse

CONSOLIDATION

Replacement of air within a lung region by infection, fluid, or tumor

INFILTRATION

Accumulated matter or cells in the lung appearing as opacities of varied size and intensity

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EDEMA

Broadly describes swelling or fluid accumulation

CARDIOMEGALY

Enlargement of the cardiac silhouette

FIBROSIS

Progressive destruction of the airways, increasing susceptibility to lung infections

HERNIA

Protrusion of lung tissue beyond the anatomical boundaries of the ribcage

PNEUMONIA

Acute deep infection of the lung parenchyma and the leading cause of childhood death globally

5. THE METHOD

Several design considerations informed the building of the dataset manifest:

DATA ACQUISITION

12 image archives downloaded and explored in a reproducible workflow and metadata labels validated

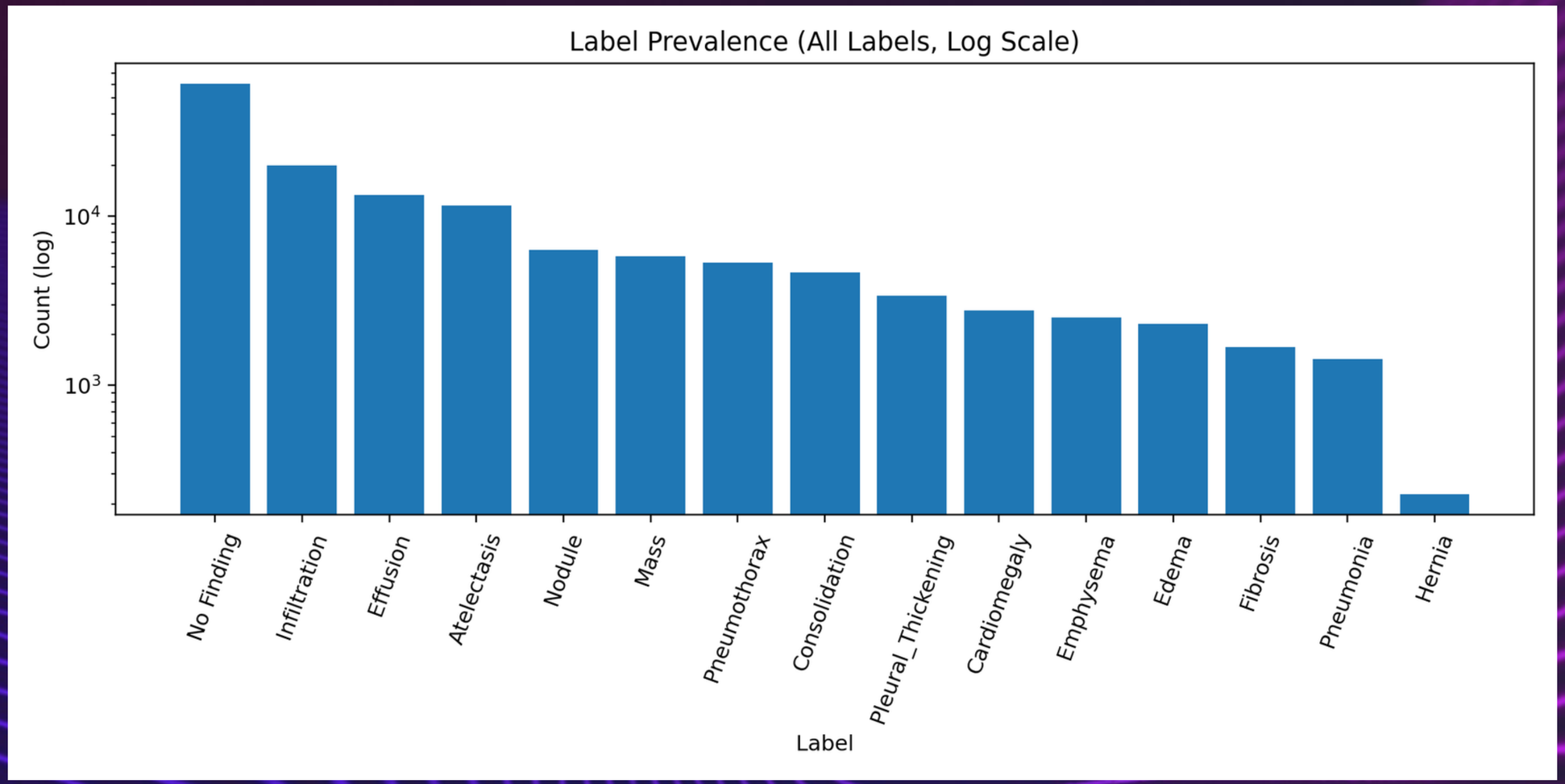
PREPROCESSING

Dataset integrity preserved by not imputing missing ages, avoiding resampling for class imbalance, keeping duplicate patients, and deferring global image normalization

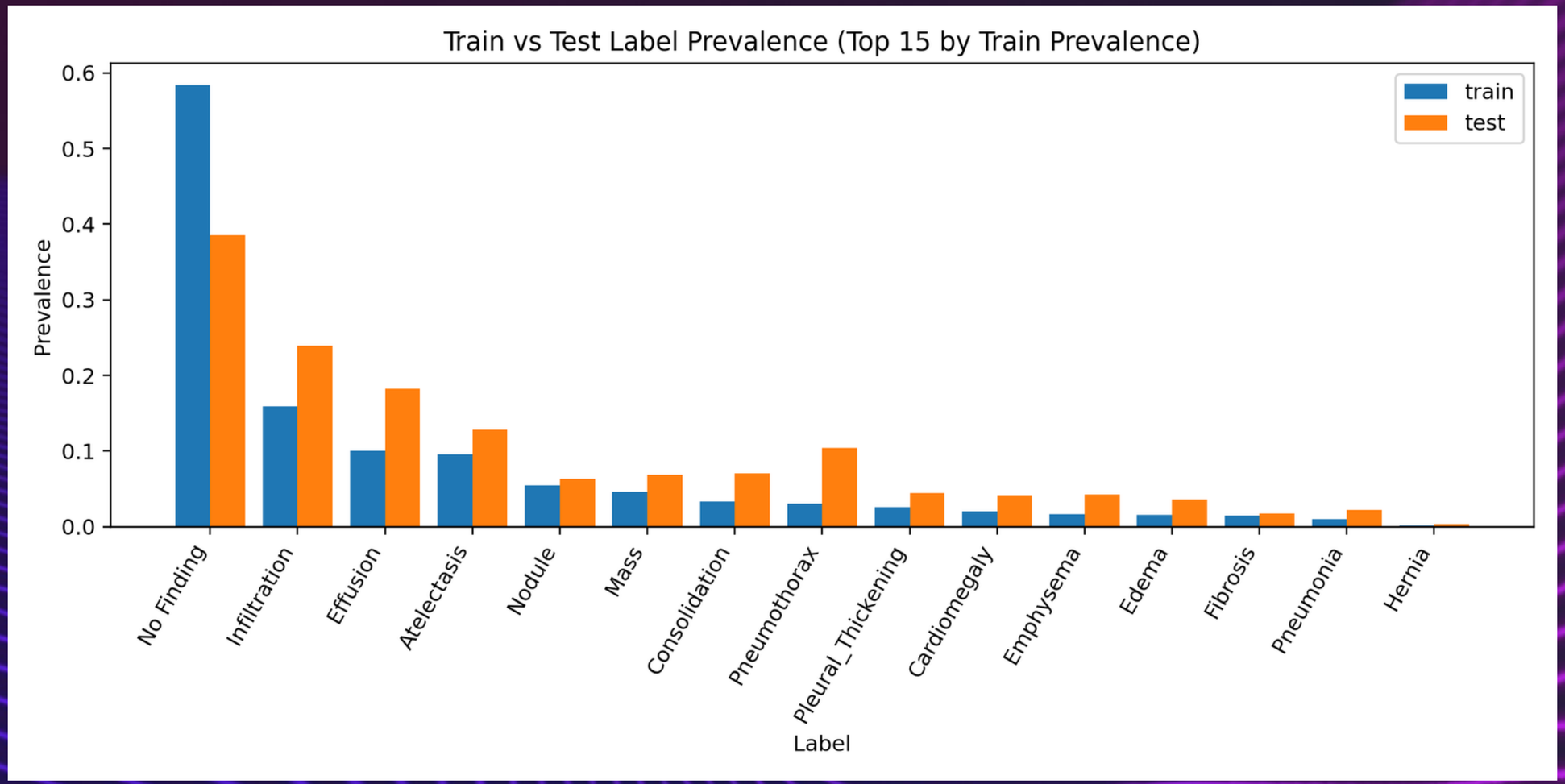
ANALYTIC METHODS

Multi-Label Representation
Co-occurrence and Dependence
Metadata Stratification
Patient-Level Leakage
Image Intensity Diagnostics

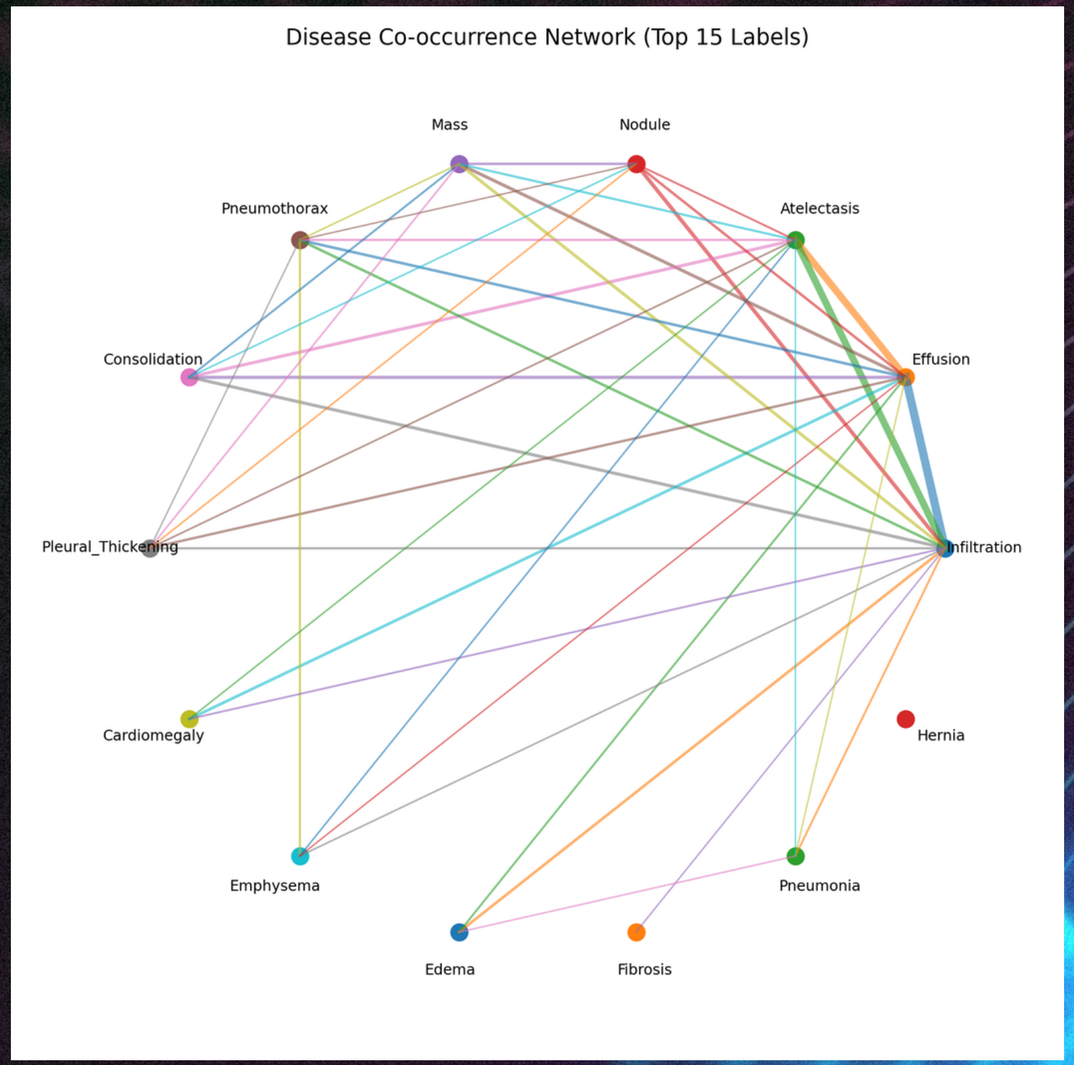
6.1. THE IMBALANCE PROBLEM

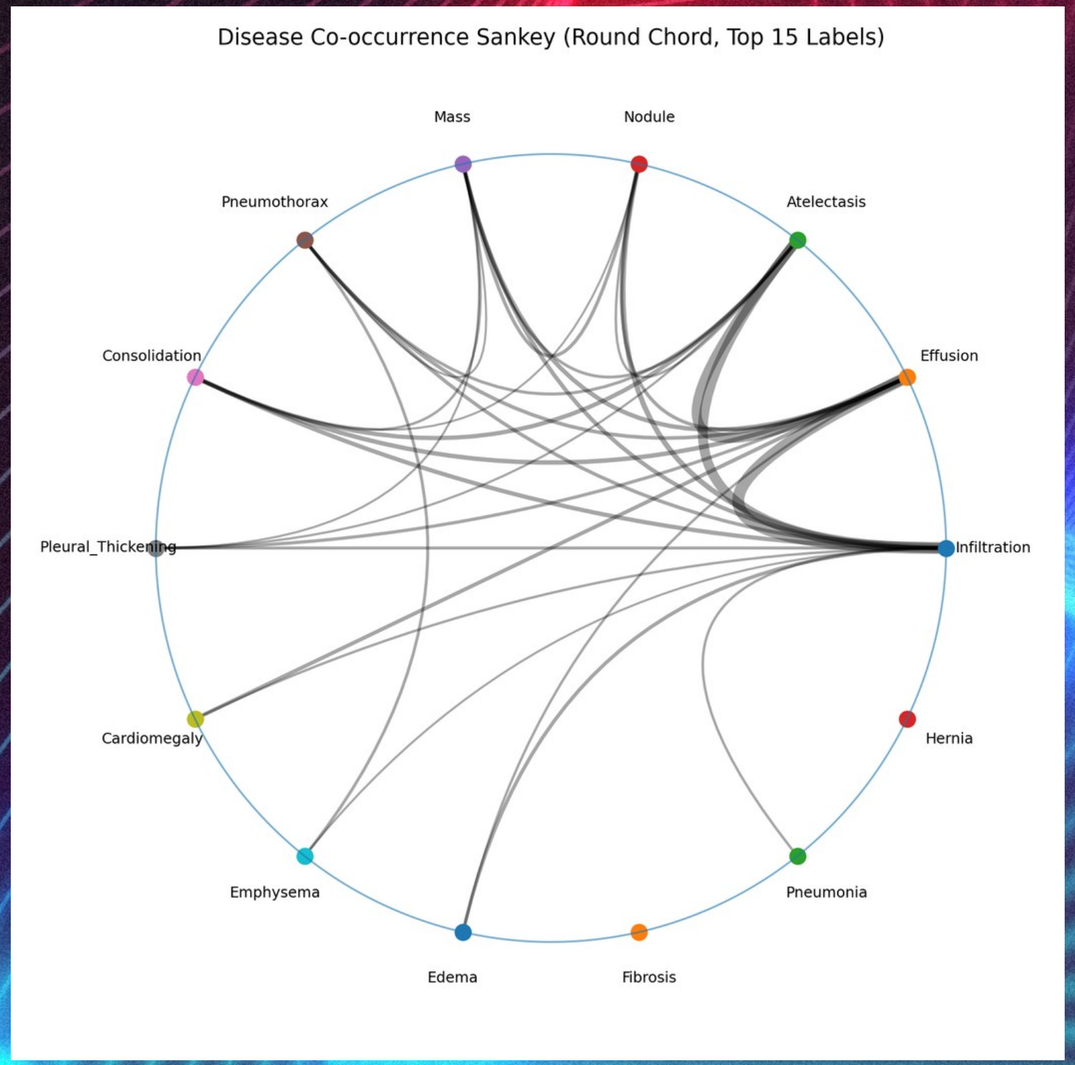


6.2. THE IMBALANCE PROBLEM

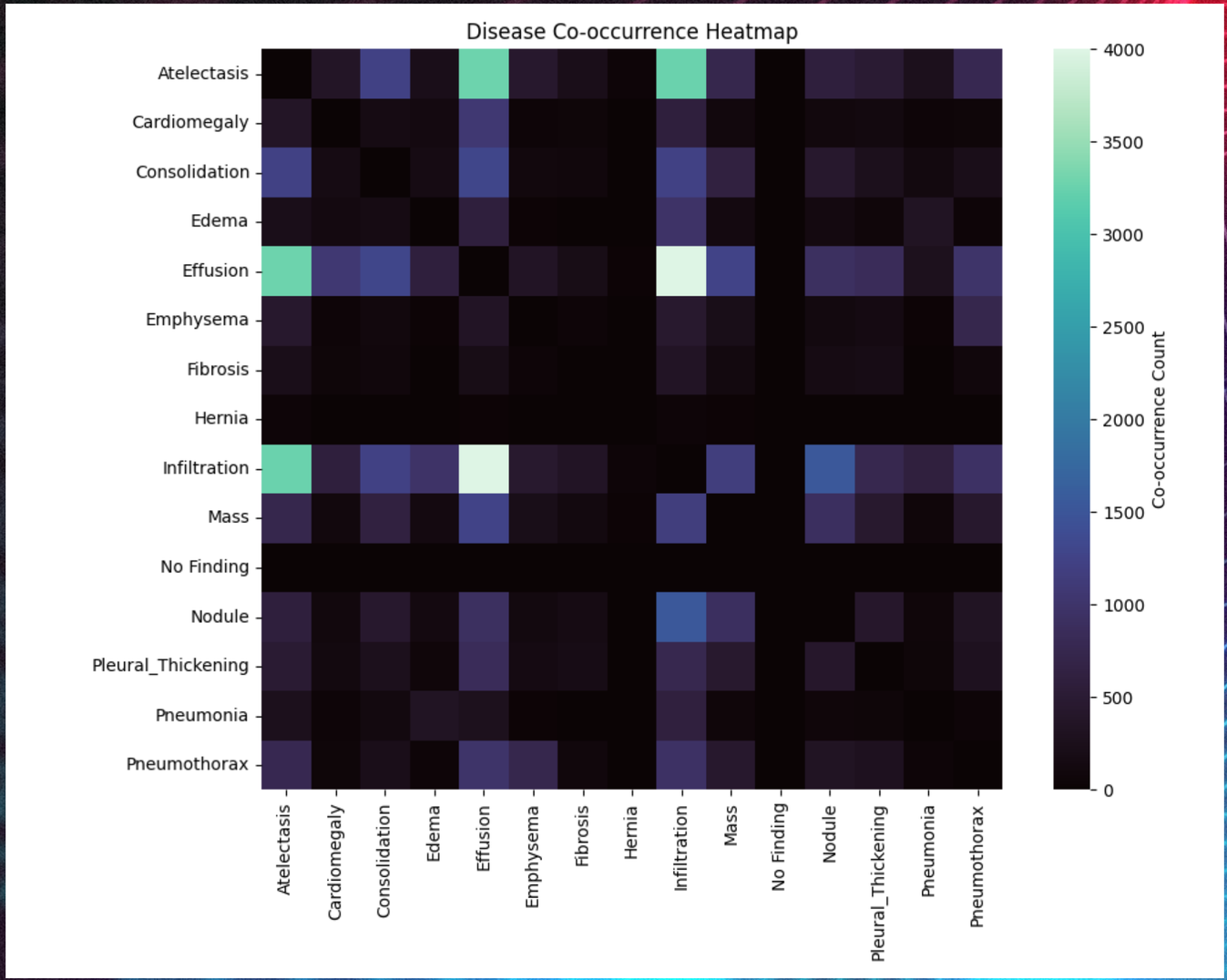


7.1. THE LABEL STRUCTURE

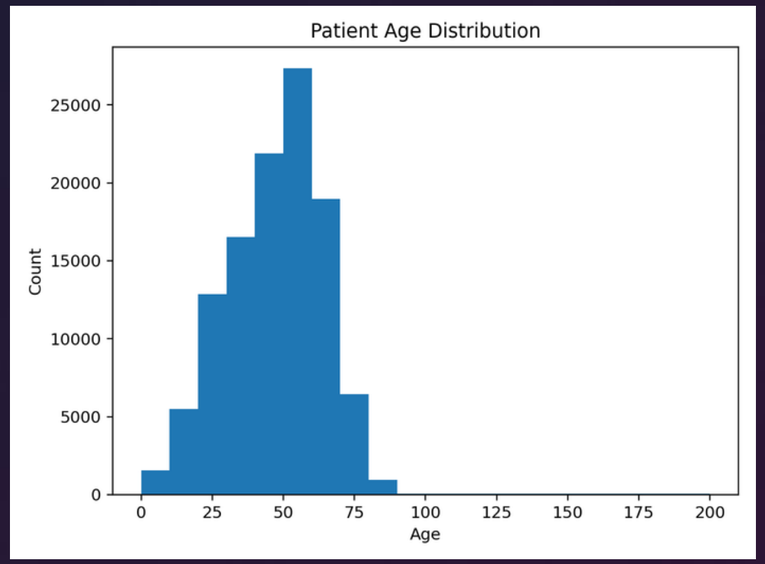


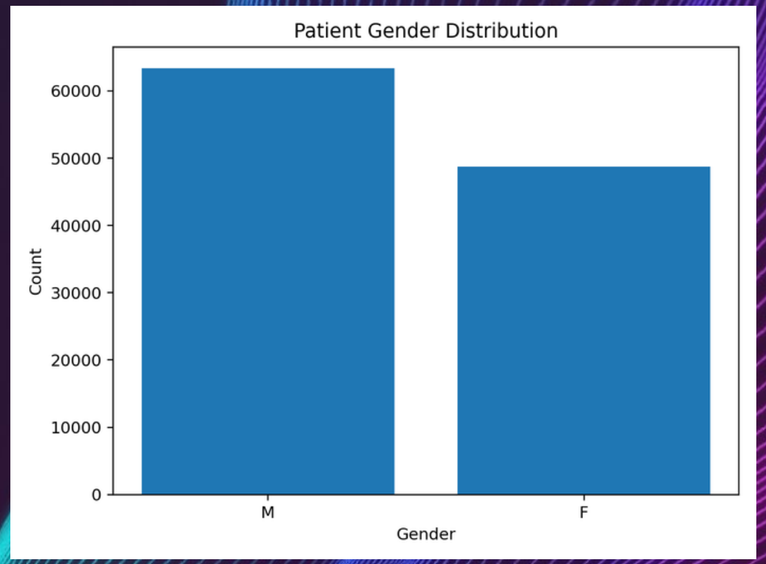


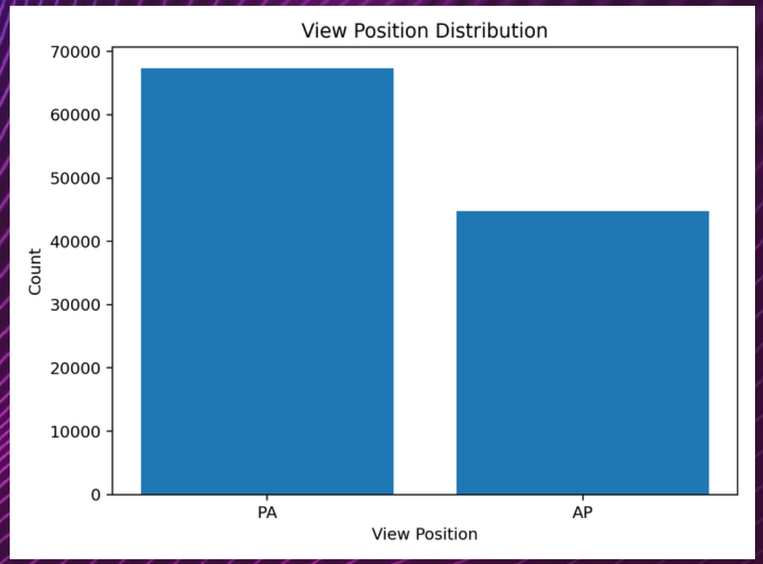
7.2. THE LABEL STRUCTURE



8.1. THE PATIENTS





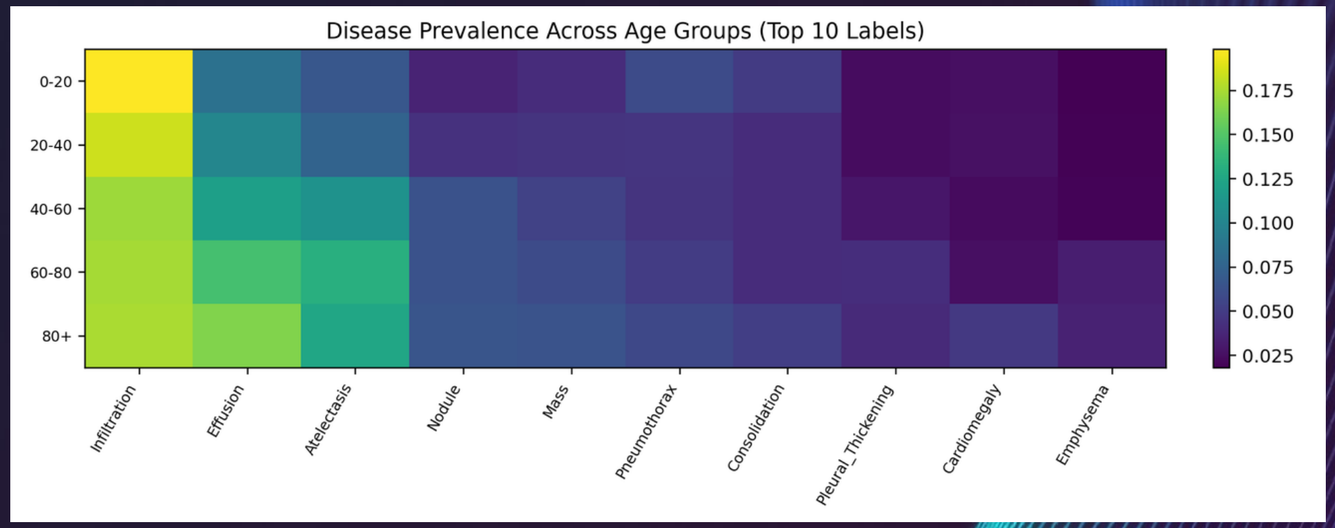


AGE

GENDER

VIEW POSITION

8.2. THE PATIENTS



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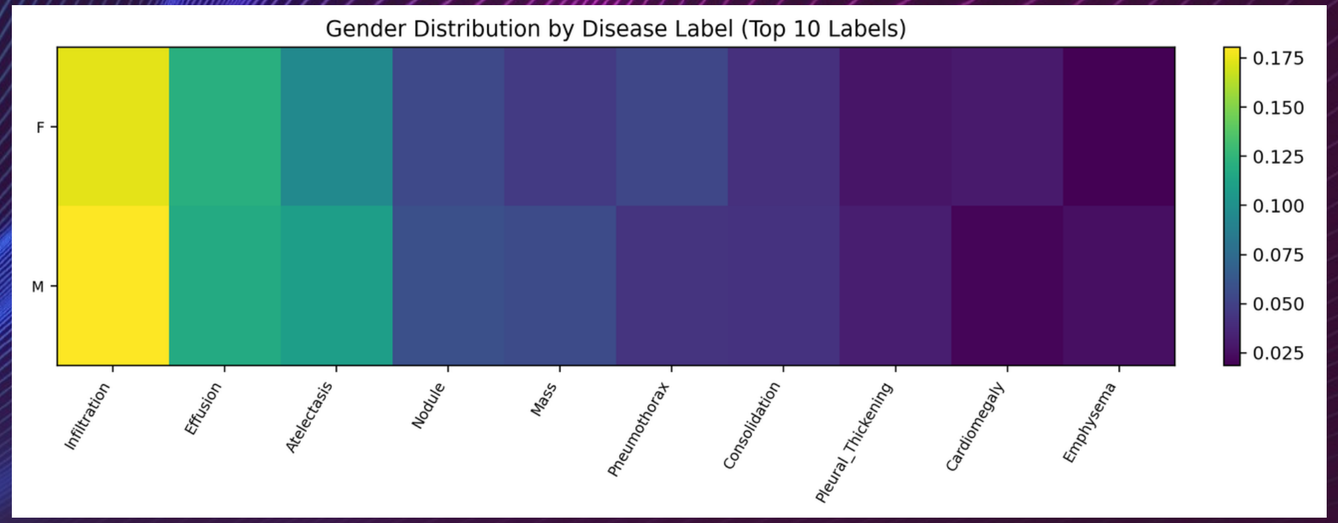


9. THE LEAKAGE AUDIT

28,008

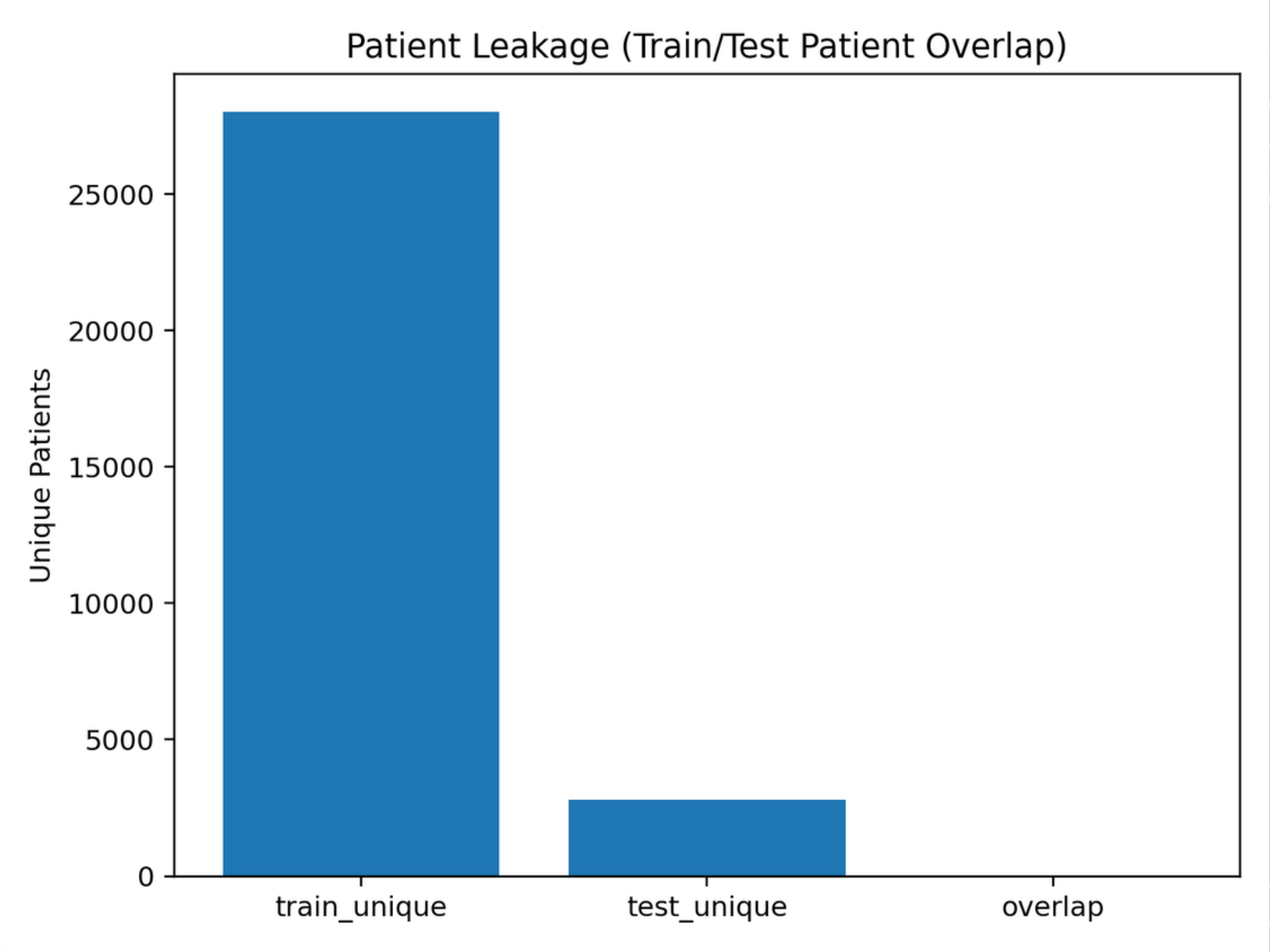
Patients in Train

2,797

Patients in Test

0%

Patient Overlap



10. THE SEQUEL

MARCH

Multi-Label Loss & 'No Finding' Handling:

Research binary cross-entropy with class-frequency inverse weights to address the 87.6:1 imbalance ratio. Consider implications of treating 'No Finding' as a separate binary indicator due to its definitional negative correlation with all disease labels.

APRIL

Explore Transfer Learning:

DenseNet-121 pretrained on ImageNet, fine-tuned on the feasible 13-label subset. ResNet-50 is included as a reference architecture to provide a standardized performance baseline and to probe data quality independently of model depth.

MAY

Model Testing & Evaluation:

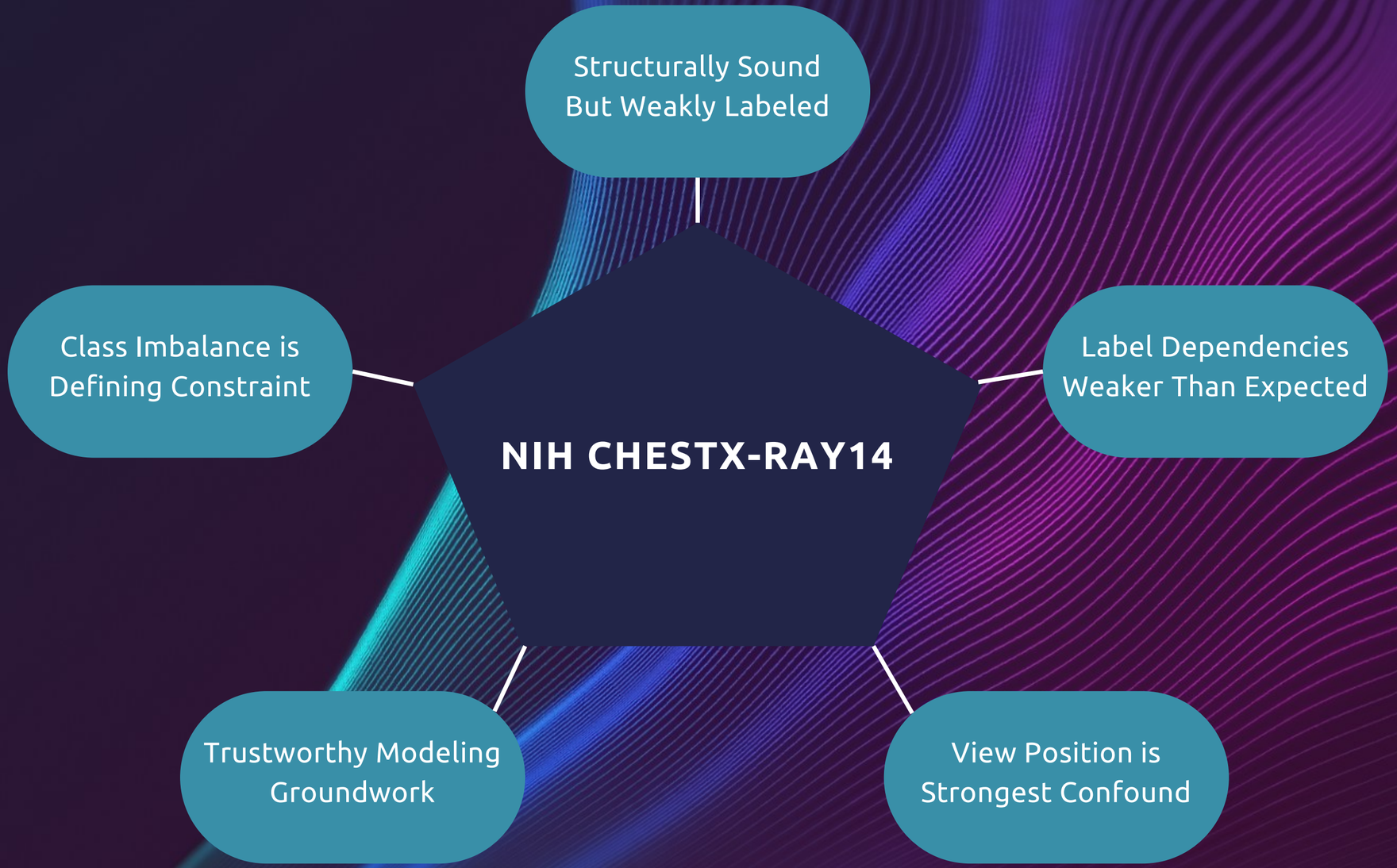
Perform model training and testing. Use AUC-ROC per label (primary metric), with subgroup AUCs stratified by gender, age bin, and view position to detect hidden stratification failures.

JUNE

Final Reporting:

Conclude testing and evaluations on model. Compile analyses and conclusions formulated, and document within final report preparation.

11. THE CONCLUSION



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THANK YOU

FOR YOUR ATTENTION

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